PODDAR
PIPES

PODDAR
UGD GOLD

PIPES AND FITTINGS

Smart and efficient drainage system

WHAT'S INSIDE?

INDEX

A LEGACY OF QUALITY

ABOUT
PODDAR PIPES

Poddar Pipes are designed to support and enhance your infrastructure, providing solid foundations for the most demanding projects. Our innovative piping solutions are trusted by engineers, builders, and architects alike to ensure seamless integration and optimal performance. Choose Poddar for pipes that reinforce your vision and elevate your project.

STANDARDS WE STAND BY

OUR COMMITMENT TO **QUALITY CERTIFICATIONS**

PERFORMANCE YOU CAN DEPEND ON

WHY CHOOSE
PODDAR PIPES

WITHSTANDS PRESSURE

Poddar Pipes are engineered to endure high pressure, ensuring long-lasting reliability and performance.

WIDE RANGE

Explore our extensive range of pipes, designed to meet diverse needs across industries.

CERTIFIED MATERIAL

Made from certified materials, Poddar Pipes guarantee safety, strength, and durability.

OUR COMMITMENT TO YOU

BUILT ON TRUST AND **ASSURANCES**

Every pipe and fitting that leaves our facility carries with it a promise — a promise of quality, precision, and performance that you can rely on. The following assurances reflect the standards, technologies, and practices that define how Poddar Pipes operates, and why professionals across the industry continue to choose us.

PRECISION MANUFACTURING LINES

Manufactured on state-of-the-art extrusion lines that are designed and maintained to deliver consistent product quality, precise dimensional accuracy, and dependable long-term performance across every single unit produced.

100% VIRGIN RAW MATERIALS

Made exclusively using 100% virgin raw materials, ensuring that every pipe and fitting delivers superior structural strength, enhanced durability, and a level of product consistency that recycled or inferior materials simply cannot match.

INTERNATIONAL QUALITY STANDARDS

All pipes and fittings are manufactured in strict compliance with IS 16098 and EN 1401 standards, providing customers with the assurance that every product meets nationally and internationally recognised quality benchmarks.

BATCH-BY-BATCH DIMENSIONAL VERIFICATION

Every production batch undergoes thorough dimensional checks and verification procedures before dispatch, ensuring that all products conform precisely to the required specifications and applicable standards before reaching the customer.

TESTED FOR UNDERGROUND STRENGTH

Ring stiffness is comprehensively tested on all relevant products to confirm reliable underground performance, resistance to external loading, and the long-term structural stability required for subsurface installations.

LIGHTWEIGHT YET STRUCTURALLY RIGID

Advanced foam core technology is employed to enhance overall pipe rigidity and structural integrity, while simultaneously reducing the total weight of the pipe for significantly easier on-site handling and faster installation.

GUARANTEED LEAK-PROOF JOINTS

Fully leak-proof joints are consistently achieved through the use of pre-fitted Black Seal™ rubber rings, which are engineered to deliver a secure, dependable, and long-lasting sealing performance across all connection points.

ISI CERTIFIED, INDUSTRY COMPLIANT

All ISI certified products are manufactured in full compliance with applicable government and industry standards, ensuring that every product meets the required benchmarks for quality, operational safety, and long-term reliability.

ITALIAN TECHNOLOGY, GLOBAL EXPERTISE

Manufactured using advanced European manufacturing technology and backed by decades of proven industry expertise, our products are engineered to bring world-class piping solutions and global standards to the Indian market.

VERSATILE RANGE, ENDLESS APPLICATIONS

Available across a comprehensive range of sizes and compatible fittings, our products are designed to meet the diverse requirements of both residential and commercial applications with equal ease and reliability.

DRAINAGE SOLUTIONS

UNDERGROUND DRAINAGE SYSTEM

Poddar's Underground Drainage System is specially designed to facilitate the safe and efficient conveyance of soil, waste, rainwater, and industrial effluents from buildings and facilities to municipal sewer networks and drainage channels. The system ensures seamless transportation of wastewater through underground pipelines, directing it towards treatment plants or approved discharge locations. Built with high-quality materials and precision engineering, it delivers excellent durability, leak-proof performance, and long service life, making it an ideal choice for residential, commercial, industrial, and infrastructure developments.

TECHNICAL DETAILS

ABOUT uPVC

uPVC (Unplasticized Polyvinyl Chloride) is a versatile and high-performance thermoplastic material that has been extensively used in plumbing, drainage, and piping systems for several decades. Its exceptional combination of strength, durability, and ease of processing makes it an ideal material for manufacturing rigid pipes and fittings. One of the key advantages of uPVC is its smooth internal surface, which promotes efficient fluid flow while minimizing friction losses and the accumulation of deposits. In addition, its robust structure allows pipes and fittings to maintain their integrity and performance over long periods of service.

uPVC is also highly resistant to corrosion, chemical attack, and environmental stress, making it suitable for a wide range of underground and above-ground applications. Its lightweight nature simplifies transportation and installation, reducing labor requirements and project costs. Furthermore, uPVC systems require minimal maintenance, ensuring reliable and cost-effective performance throughout their operational lifespan.

PHYSICAL PROPERTIES

Density [g/cm3]	1.3 - 1.45
Thermal conductivity [w/(m.k)]	0.14 - 0.28
Yield strength [MPa]	31 - 60
Young's modulus [psi]	4,90,000
Flexural strength (yield) [psi]	10,500
Compression strength [psi]	9,500
Coefficient of thermal expansion (linear) [mm (mm"c)]	5×10^{-5}
Vicat B [°c]	65 - 100
Resistivity [Ωm]	10^{16}
Surface resistivity [Ω]	$10^{13} - 10^{14}$

TECHNICAL DETAILS

ABOUT
FOAM CORE

Underground drainage systems comprise pipes, fittings, and accessories designed to convey household sewage, rainwater, and industrial effluents safely and efficiently. These systems are commonly installed between buildings and municipal sewer networks and operate under non-pressure conditions. Depending on application requirements, underground drainage systems can be manufactured from various materials such as PVC, HDPE, clay, concrete, and fiberglass.

Poddar Underground Drainage Systems are manufactured using high-quality uPVC, a material widely recognized for its durability, reliability, and long service life. To further enhance performance, Poddar utilizes a three-layer foam-core pipe construction, consisting of solid uPVC outer and inner layers with a lightweight foamed uPVC core. This advanced structure improves stiffness while reducing pipe weight, resulting in easier handling, efficient installation, and superior underground performance.

KEY ADVANTAGES OF uPVC

- High flow efficiency due to smooth internal surfaces
- Excellent thermal and electrical insulation properties
- Strong yet lightweight construction
- Superior resistance to chemical stress cracking
- Outstanding chemical stability and corrosion resistance
- Leak-proof jointing system for reliable performance
- Safe and environmentally compatible material for drainage application

ADVANTAGES OF FOAM CORE

Foam core pipes are basically multi-layer pipes having outer and inner layers of conventional uPVC and a middle layer of foamed uPVC. The outer and inner layers are designed to withstand external loads while providing excellent chemical resistance, whereas the foamed core enhances rigidity and helps maintain the pipe's shape under pressure. Due to their load-absorbing capability, foam core pipes are highly suitable for underground drainage systems where surrounding soil exerts continuous pressure on pipe surfaces.

In addition, foam core pipes are significantly lighter than solid-wall pipes, making transportation, handling, and installation easier and more economical. Their optimized structure reduces material consumption without compromising mechanical performance. The superior stiffness-to-weight ratio enables long-term durability, while the smooth internal surface ensures efficient flow characteristics. These features make foam core pipes a reliable, cost-effective, and sustainable solution for modern underground drainage infrastructure.

- Light weight and strong
- High impact strength
- Water ingress prevention
- Thermal insulation
- Wide temperature range
- Dampening of vibrations and noise level

WHY IT STANDS APART

WHY PODDAR'S UNDERGROUND DRAINAGE SYSTEM?

Poddar provides a comprehensive range of underground and surface drainage systems designed to meet industry standards. These solutions ensure exceptional installation flexibility, premium quality finish, outstanding dimensional precision and stability, making them ideal for both residential and commercial applications.

These pipes are manufactured using advanced extrusion technology and socketed through inline belling equipment. The fittings are produced with collapsible core moulds to ensure precise grooves and excellent dimensional accuracy, delivering enhanced structural strength. Poddar's underground drainage systems provide outstanding durability and finish, offering a leak-resistant, maintenance-free, and long-lasting drainage solution.

Poddar's underground drainage pipes are available in 110 mm to 315 mm in Pushfit and Solfit jointing methods with different stiffness classes mainly categorised as SN2, SN4 and SN8.

- 110 mm with stiffness class SN4 and SN8
- 160 mm, 200 mm, 250 mm and 315 mm with stiffness class SN2, SN4 and SN8 VV

WIDE RANGE OF PIPES AND FITTINGS

Sizes available for Pipes and fittings in Pushfit and Solfit – 110, 160, 200, 250 and 315 mm.

FULLY ANALYZED RAW MATERIALS

The raw materials used for manufacturing the products are fully analyzed and procured with utmost care to deliver high quality products to the customers.

RODENT PROOF

Smooth surfaces and the rigidity of uPVC are hindrance for rodents from damaging the system.

POWERED BY PODDAR TECHNOLOGY

Engineered for durability, precision, efficiency, and long-lasting performance across every application.

SPECIAL FITTINGS FROM PODDAR

Specialized fittings such as large-diameter non-return valves, easy clips, mechanical saddles, swivels, and select higher-diameter fittings are sourced from Poddar to complement the drainage system range

WIDE DISTRIBUTION NETWORK

Poddar has an extensive distribution network across the country, ensuring easy availability of its products through local dealers and retail outlets.

BEST MANUFACTURING STANDARDS

The pipes are produced on state-of-the-art extrusion lines and socketed using online belling machines. The fittings are manufactured with collapsible core moulds to create clean, well-defined grooves and maintain high dimensional accuracy, resulting in consistent quality and reliable performance.

OUR FEATURES

- Light weight and strong
- Wide range and compatibility
- Easy to Install
- Maximum flow rate
- Chemical and corrosion resistance
- Non toxic and non conductor
- Longer service life
- Non flammable
- Environmentak friendliness
- Noise free

WHY IT STANDS APART

WHY ONLY PUSHFITS?

Advanced moulding technology enables the production of bi-moulded rubber rings that provide greater reliability and remain securely positioned within the groove during installation. The system eliminates the need for threading or solvent cementing, while the spigot end is firmly retained in the socket by the Black Seal. This creates a leak-proof joint capable of withstanding high-pressure flow conditions. Manufactured using advanced, next-generation machinery, the system delivers

The advanced Black Seal allows joints to withstand high pressure while ensuring a reliable leak-proof connection and accommodating the thermal expansion and contraction of plastic pipes. As the joint is not permanently fixed using solvent cement, it can be reopened after installation for realignment, modification, or adjustment of the pipe and fittings whenever required.

THERMAL EXPANSION

uPVC has a thermal expansion coefficient of approximately 0.06 mm/m/°C. As a result, a 2 m length of soil or waste pipe can expand by about 2.4 mm when subjected to a temperature increase of 20°C. This movement is considered during the design of the piping system and its components and must also be accounted for during installation. To accommodate thermal expansion, an expansion gap should be provided at ring seal joints. The spigot end should be inserted fully into the ring seal socket, marked at the socket face, and then withdrawn by 10 mm to create the required expansion

THE PODDAR ADVANTAGE

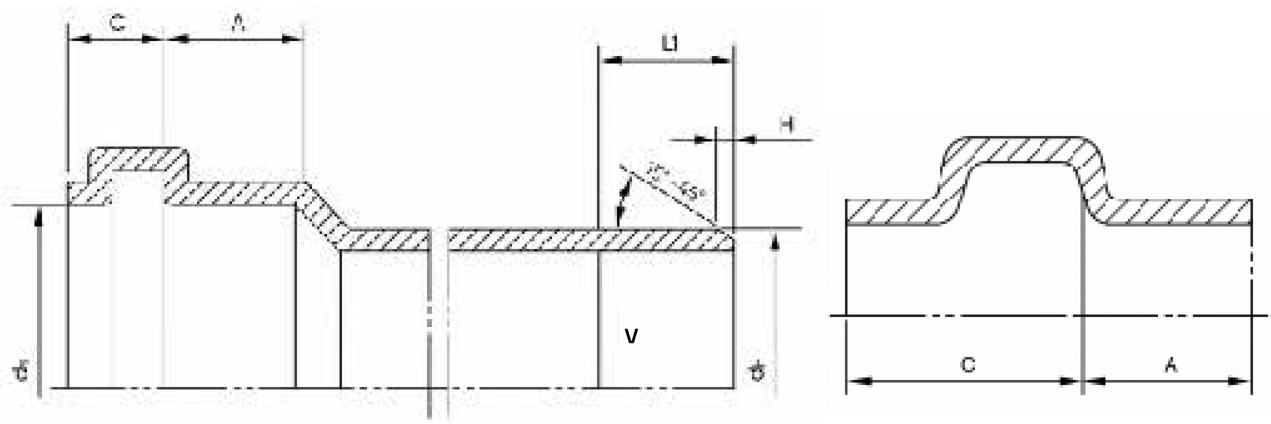
- Smooth inner finish
- Corrosion & abrasion resistance
- Tough & Reliable
- Quick & Easy installation
- Cost effective

Poddar Pipes are engineered for strength, durability, and reliable performance. With a smooth inner finish, superior corrosion and abrasion resistance, quick installation, and cost-effective operation, they deliver long-lasting efficiency for every plumbing application.

TECHNICAL SPECIFICATIONS FOR UGD PUSHFIT SYSTEM

Poddar Underground Drainage System pipes are manufactured as per IS 16098 (Part 1): 2013 standards, while the fittings are produced in accordance with EN 1401-1:2009 (SDR 41) specifications. Available in a complete range of sizes including 110, 160, 200, 250 and 315 mm, the system is designed to satisfy the drainage requirements of residential, commercial and industrial applications.

The system is supplied with a pre-fitted rubber Black Seal in the groove, ensuring a reliable leak-proof joint. Installation is simple and efficient, requiring the spigot end to be pushed into the socket end.



Typical Groove design for elastomeric ring seal sockets

DIAMETER AND LENGTHS OF ELASTOMERIC RING SEAL SOCKETS AND SPIGOTS ENDS

Nominal outiside Diameter, dn	Mean OutsideDiameter (mm)			Socket		Spigot	
	Min	Max	ds,min	A min	C max	L1, min	H
110	110	110.3	110.4	32	26	60	6
160	160	160.4	160.5	42	32	81	7
200	200	200.5	200.6	50	40	99	9
250	250	250.5	250.8	50	70	125	9
315	315	315.6	316	62	70	132	12

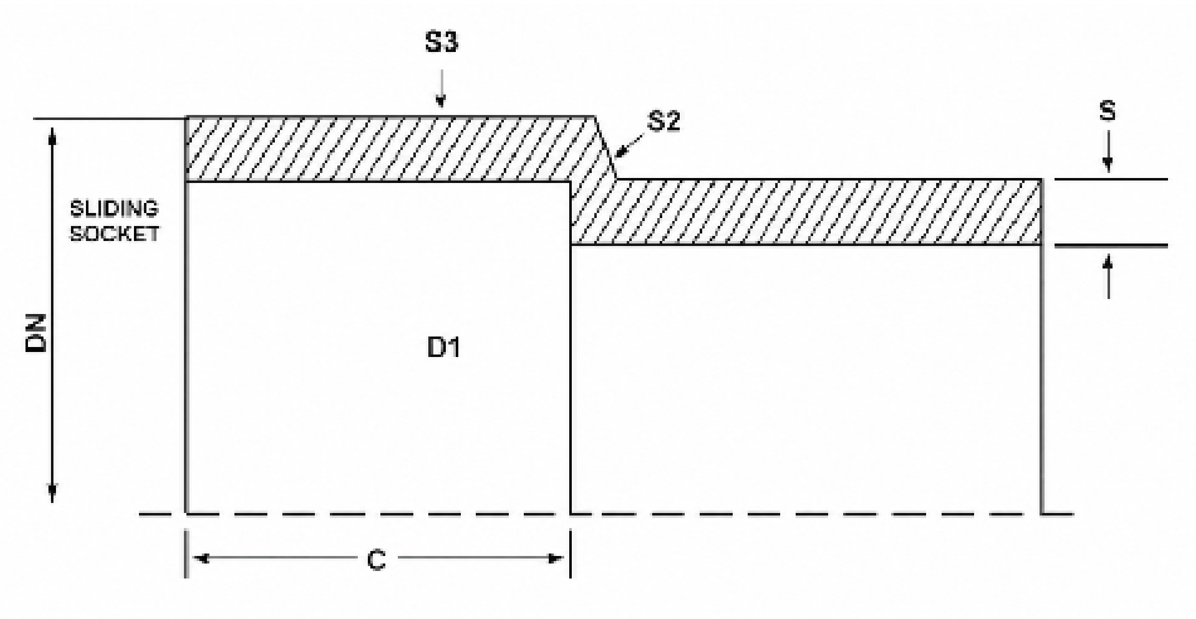
WHY IT STANDS APART

WHY ONLY SOLFIT?

Poddar Solfit Underground drainage system pipes are manufactured as per IS 16098 (Part 1) : 2013 standard and fittings are manufactured as per EN 1401- 1:2009 (SDR 41) standard in sizes ranging from 110, 160, 200, 250 and 315 mm. Poddar Solfit systems are joined by solvent adhesive. This system is made by new generation high tech machines and offers unrivalled performance, strength and finish.

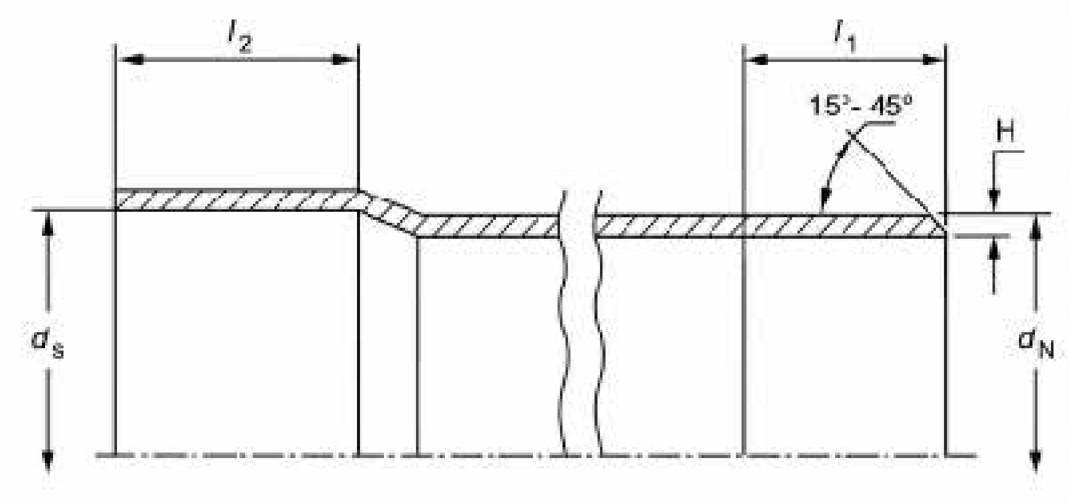
KEY PERFORMANCE BENEFITS

- High flow rates -no choking
- Cost effective
- 100% leak proof joints
- High degree of dimensional accuracy



TECHNICAL SPECIFICATIONS FOR UGD SOLFIT SYSTEM

Poddar Solfit Underground drainage system pipes are manufactured as per IS 16098 (Part 1) : 2013 standard and fittings are manufactured as per EN 1401-1:2009 (SDR 41) standard in sizes ranging from 110, 160, 200, 250 and 315 mm.



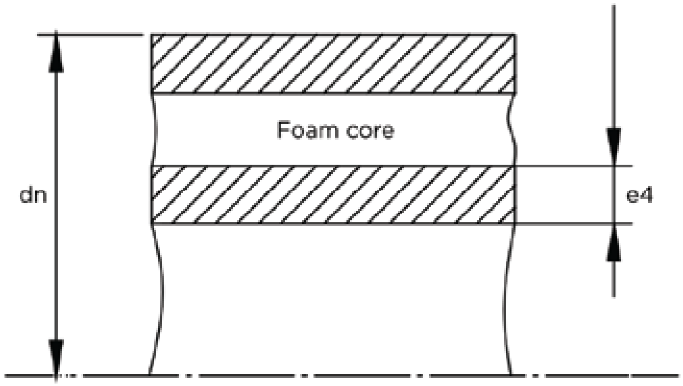
Typical Groove design for elastomeric ring seal sockets

DIMENSIONS OF SOCKETS AND SPIGOTS ENDS FOR SLIDING / SOLFIT SOCKET JOINTS

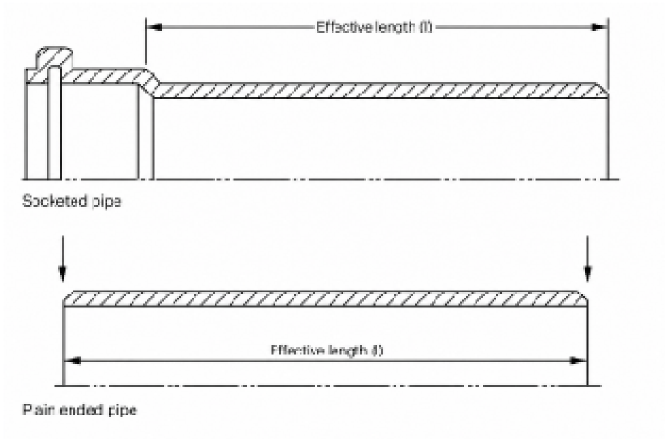
Nominal outiside Diameter, dn	Mean Outside Diameter (mm)			Socket		Spigot	
	Min	Max	ds,min	ds,max	L2, min	L1, min	H
110	110	110.3	110.1	110.4	61	71	6
160	160	160.4	160.2	160.4	86	96	7
200	200	200.5	200.3	200.6	106	116	9
250	250	250.5	250.4	250.8	131	141	9
315	315	315.6	315.4	316	163.3	193	12

TECHNICAL
SPECIFICATIONS

DIMENSION OF PIPES AS PER IS 16098 – PART 1 (2013)



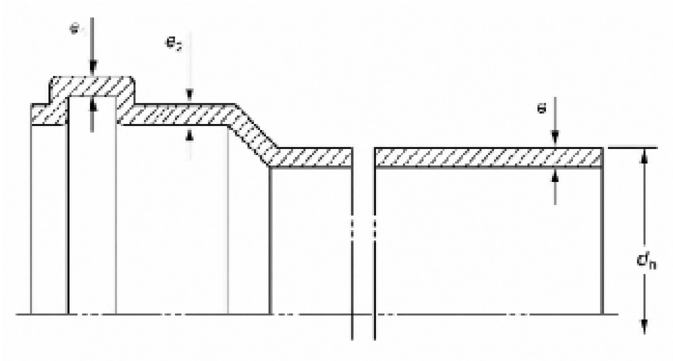
EFFECTIVE LENGTH OF PIPE



OUTSIDE DIAMETER AND THICKNESS OF INSIDE LAYER

Nominal outiside Diameter,dn	Minimum Wall thickness of inner layer (e3, Min)
110	0.4
160	0.5
200	0.6
250	0.7
315	0.8

DIMENSION OF FITTINGS AS PER EN 1401-1 : 2009 (SDR 41) STANDARDS



PIPE AVAILABILITY - PUSHFIT & SOLFIT

Description	SN2	SN3	SN4
Pipe Diameter	Availibility	Availibility	Availibility
110 mm	Not covered in IS	✓	✓
160 mm	✓	✓	✓
200 mm	✓	✓	✓
250 mm	✓	✓	✓
315 mm	✓	✓	✓
1/2	✓	✓	✓

WALL THICKNESS OF FITTINGS

Nominal outside Diametere, dn	SN 4 (SDR1)			
	Spigot		Socket	
	e min	e max	e2 min	e3 min
110	3.2	3.8	2.9	2.4
160	4.0	4.6	3.6	3.0
200	4.9	5.6	4.4	3.7
250	6.2	7.1	5.5	4.7
315	7.7	8.7	6.9	5.8

THE STRENGTH

RING STIFFNESS

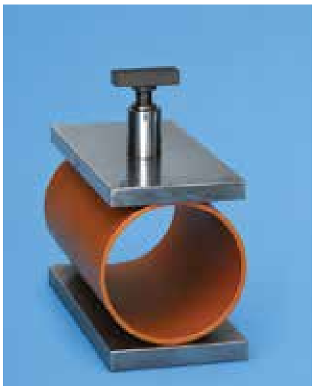
PRINCIPLE

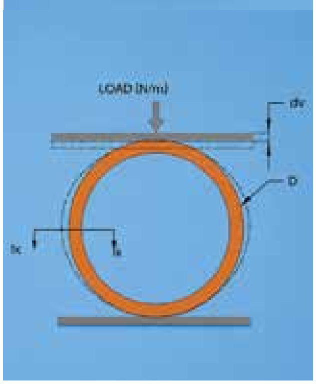
The ring stiffness of a pipe indicates its ability to resist soil loads, external hydrostatic pressure, negative internal pressures and traffic loads. Nominal ring stiffness can be determined by laboratory testing and is expressed as kN/m2.

Reference IS 16098 (Part 1) : 2013. All diameters of Poddar foamcore pipes are manufactured with a constant minimum stiffness as either designated as SN2, SN4 and SN8 respectively in accordance with the standards.

TEST PROCEDURE

The ring stiffness is determined by measuring the force and the deflection while deflecting the pipe at a constant rate. A length of pipe supported horizontally is compressed vertically between two parallel flat plates moved at a constant speed, which is dependent upon the diameter of the pipe. A plot of force versus deflection is generated. The ring stiffness is calculated as a function of the force necessary to produce a deflection of 0.03D diametrically across the pipe.





THE PIPE IS SAID TO HAVE PASSED THE TEST IF :

- There is no cracking or crazing of the inside wall or liner
- There is no wall delamination
- The test piece has not ruptured and there are no mechanical failures
- There is no change in shape of cross section of the pipe (buckling)

TABLE 1 – RING STIFFNESS OF PIPES

Stiffness Class	Ring Stiffness (kN/m™)
SN2	≥ 2
SN4	≥ 4
SN8	≥ 8

MANUFACTURING STANDARDS

QUALITY CONTROL PROCEDURES

Pipes and fittings manufactured at Poddar undergo a stringent quality control process before being released into the market, ensuring the supply of a reliable and defect free drainage system. Every stage of manufacturing is carefully monitored to maintain consistent quality, dimensional accuracy and performance. Strict adherence to standards and thorough inspections ensure durability, reliability and long service life, delivering dependable piping solutions for a wide range of applications.

PIPES

DIMENSIONAL CHECK

To ensure that all pipe dimensions, particularly wall thickness and outer diameter (roundness), conform to the appropriate standards.

LONGITUDINAL REVERSION

A pipe section is heated at a specified temperature for a set time. The marked length is measured before and after heating, and reversion is calculated as the percentage change from the original length.

RING STIFFNESS TEST

The stiffness of a pipe indicates the ability of pipe to resist the external soil, hydrostatic and traffic loads together with negative internal pressure.

RING FLEXIBILITY TEST

Ring flexibility is determined by measuring the force and deflection while deflecting a ring section from the pipe diameter at a constant speed until a deflection of at least 30 percent.

FITTINGS

DIMENSIONS CHECK

To ensure that fittings have correct dimensions, particularly wall thickness, socket diameters and socket depth.

DROP IMPACT TEST

Fittings are dropped at 0°C from 1 meter height on the floor and observed for any cracks or failures.

OTHER TESTS

- Color
- Vicat softening temperature
- Resistance to dichloromethane
- Creep ratio
- Resistance to combined temperature cycling and external loading
- Tightness of elastomeric sealing ring joints at 2° deflection
- Stress relief test

SPECALITY FITTINGS

MECHANICAL CONNECTIONS

Poddar introduces high quality mechanical connections that can be used to join underground drainage pipes to municipal sewers. These products can be used to join to sewers made out of PE/PP/PVC - solid wall, corrugated, clay, concrete, GRP or cast iron (thickness 5 mm to 14 mm, diameter 200 mm to 40 mm).

EASY CLIP (160 MM AND 200 MM DIA)

Easy clip is used for the creation of sewer line connections using a watertight coupling device with a mechanical anchoring system. The body of the clip is made from PVC and seals are made from EPDM. Easy clip fit for purpose saddles have been successfully installed for conveying surface waters from highway bridges and to connect private underground drainage systems to the main sewers.These products can be used to join to sewers made out of PE/PP/PVC - solid wall, corrugated, clay, concrete, GRP or cast iron (thickness 5 mm to 100 mm, diameter 300 mm to 1200 mm).

INSTALLATION STEPS

Identify the point at which to perforate the pipeline and drill the hole perpendicular to the axis of the main pipe.

Use a suitable tool to smooth the edge of the hole as well as lubricate the seal. Also ensure a good fit of the circular interface.

Insert the saddle and tighten the ring nut to compress the seal and create a leak-proof joint.

ADVANTAGES OF INSTALLING EASY CLIP

- Short excavation time
- Soil re-compaction time
- No alteration of bedding
- No risk of rupture
- No digging beneath the pipeline
- Speed of installation
- Flexible solutions
- Ease of installation
- No utility interference
- No sealants required

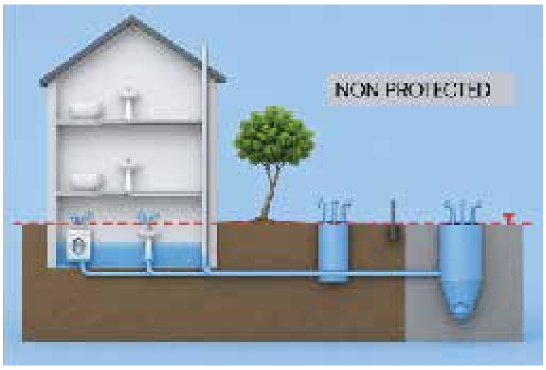
BACK FLOW PREVENTER / NON RETURN VALVES

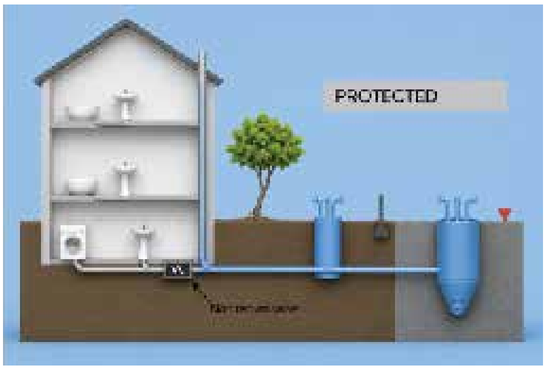
Non return valves are simple and effective devices for reducing the risk of flood damage caused by surface water flowback through the underground drainage system. Injection moulded non return valves made of uPVC designed for connections to piping systems conform to European EN 1329 and EN 1401 standards.

THE MAIN CAUSES OF BACK FLOW ARE

- Overall under-sizing of the public sewage.
- High peak of flow (example in case of intense rainfalls).
- Urban growth demands higher sewage evacuation.
- Malfunctioning or blockages downstream in the network.

MAIN FUNCTIONING COMPONENTS AND OPERATION





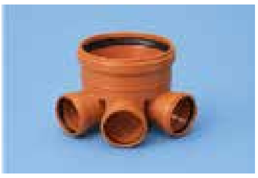
INSTALLATION

Installation of NRV is within an access point between the facility to be protected and main sewer. Any flowback will automatically shut the valve. This product can be obtained in sizes from 110 mm to 315 mm from Poddar.

SWIVEL

Swivel joint for the compensation of settling side when used with inspectionchambers. Special kneecap compensation whose seat is obtained by the coupling of 2 hemispheres. The product comes from PODDAR mounted, pre-lubricated and ready to use. The end can absorb inclinations of the pipe up to +/- 10°. Installation: Suitable for systems with O-ring seal. Always chamfer cut pipe and lubricate all plain ended spigots for perfect push-fit.

INSTALLATION PROCEDURE



DN 160 MM

- Pre lubricated and ready to install
- Rotation up to 10° to ease house connections and balance ground settlement

WATER INTELLIGENCE

SMART WATER MANAGEMENT

With technical tie ups from across the globe, Poddar strives to bring thve latest technology and products into the Indian plumbing market, with more and more satisfied customers each day.

APPLICATION OF UNDERGROUND DRAINAGE SYSTEM

Poddar underground drainage and sewerage system has been designed with a view to modern man's inclination towards health, hygiene and his aversion to filth and pollutants.
Due to unpleasant nature of human waste, a drainage system should be “out of sightand out of mind”. Most of the drainage systems are actually hidden from sight.

Poddar underground system can also be used for rainwater collection and disposal, including rainwater harvesting. In short, the system provides complete solution for underground drainage and sewerage applications.

HANDLING AND STORAGE

PROPER HANDLING

- Please check and inspect the pipes on receipt. The pipes should be checked for any forms of transport damage due to shift in loads or improper handling/treatment. Visually examine the ends of pipes for any cracks or damage.
- The pipes should be handled with care. The tendency to throw or drop the pipes to the floor should be avoided. Do not drag or push the pipes from a truck bed. Contact of pipes with from any sharp object should be totally avoided.

STORAGE OF PIPES

- The pipes should preferably be stored indoors. When this is not possible, please ensure
- Protect the pipes from sun light, to reduce the effect of UV rays.
- Store on level ground and dry surface.
- If pipes of same diameter but different classes are being stacked together, place the thicker pipes below. i.e., Stack below SN2, SN4 and SN8. If placing pipes on racks, ensure the spacing between the supports does not exceed 3 feet.

CORRECT TRANSPORTATION PROCEDURE

- Where possible use a truck for deliveries. Lay pipe flat on the tray.
- Keep pipe strapped down so it doesn't roll around and remains supported.
- Alternate socket and pipe ends when loading pipe.

STEP-BY-STEP

INSTALLATION GUIDE FOR PUSHFIT

Easy & 100% leakproof installation.

STEP 1

CUTTING

Measure and cut the pipe squarely to ensure proper solvent cement bonding. Inspect the pipe end for cracks or splits and remove at least 25 mm beyond any visible damage before use.

STEP 2

DEBURRING/BEVELING

Burrs in and on pipe end can obstruct flow/proper contact betweenthe pipe and socket of the fitting during assembly and should be removed from both in and outside of the pipe. A 15 mm dia half round file/a pen knife or a Deburring tool are suitable for this purpose.

STEP 3

FITTING PREPARATION

Use a clean, dry cloth to thoroughly wipe away all dirt, dust, and moisture from the fitting and the pipe end before installation, ensuring a clean surface.

STEP 4

CHECK FOR BLACK SEAL

Check the socket end for Black Seal . Ensure that the brown part of the seal is towards the outside of socket

STEP 5

LUBRICANT

Apply the recommended lubricant evenly on the chamfered end of the pipe to ensure smooth insertion, reduce friction, and achieve a secure, leak-proof joint during installation.

STEP 6

ASSEMBLY

Insert the pipe into the fitting socket and rotate slightly. Pull back until the mark is 12 mm from the socket, creating an expansion gap to prevent joint distortion.

STEP-BY-STEP

INSTALLATION GUIDE FOR SOLFIT

Easy & 100% leakproof installation.

STEP 1

CUTTING

Measure and mark the pipe accurately using a felt-tip pen. Ensure pipe and fittings are size-compatible. Cut the pipe squarely (90°) using a suitable cutter to ensure optimal bonding. Inspect pipe ends thoroughly and remove at least 25 mm beyond any visible crack or splinter before proceeding.

STEP 2

DEBURRING/BEVELING

Remove burrs from the inside and outside of the pipe end to ensure proper flow and joint fit. Use a file, knife, or deburring tool, and add a slight bevel to the pipe end for easier insertion into the fitting socket.

STEP 3

FITTING PREPARATION

Using a clean dry rag, wipe the dirt and moisture from the fitting sockets and pipe end. Dry fit the pipe to ensure total entry into the bottom of the fittings socket and make a visible marking using a felt tip pen.

STEP 4

SOLVENT ADHEISIVE APPLICATION

Apply an even coat of solvent adhesive on the pipe and the socket end of the fitting. Do not use thickened or lumpy solvent adhesive. It should have a flow consistency like that of syrup or paint

STEP 5

ASSEMBLY

Immediately insert the pipe into the fitting socket, rotate the pipe ¼ to ½ turn while inserting. This motion ensures an even distribution of adhesive within the joint. Hold the assembly for 10 seconds to allow the joint to setup.

TRENCH EXCAVATION

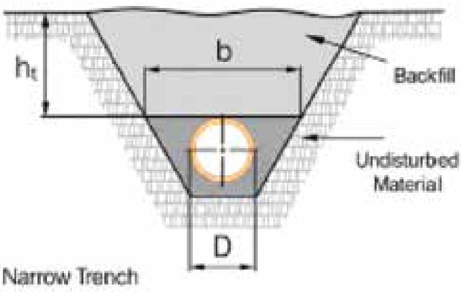
Trench excavation are carried out principally to allow installation or repair of public utilities, drains and sewers to serve populated areas The density of the backfill material, the width of the trench at the crown of the pipe, b, and the nominal outside diameter of the pipe, D, and the height of backfill (ht) all influence the loads imposed on the pipe.

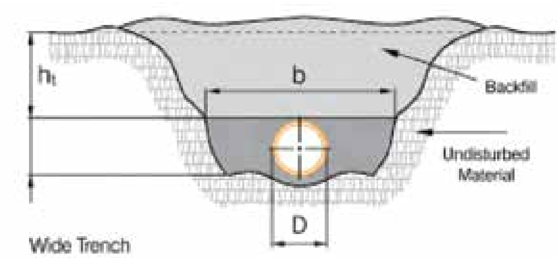
TYPES OF TRENCH

NARROW TRENCH

The pipe is laid in a relatively narrow trench with backfill properly compacted, the weight of fill is jointly supported by both the pipe and the frictional forces along the trench walls.

The narrow trench condition is used where excavation commences from the natural ground surface without any fills above the surface.





WIDE TRENCH

A wide trench gives rise to loads greater than those which occur in narrow trench owing to the greater mass of backfill bearing on the pipe, although friction between the undisturbed trench side and the backfill reduces the load to some extent.

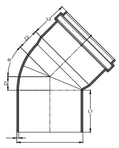
RECOMMENDED STEPS FOR INSTALLATION

- Trench excavation
- Compaction
- Sand bedding and pipe laying
- Sand filling upto crown of the pipes
- Side filling
- Back filling
- Final compaction

CAUTION

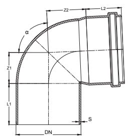
When a trench excavation is being planned for, a thorough search should be made for information about any existing utilities adjacent to or crossing the line of the planned trench, including their sizes, locations and alignments. Most of the underground utilities are live systems, such as electricity, water, sewer and gas, and can be dangerous to workers when damaged or fractured.

PUSHTIFT TECHNICAL DETAILS

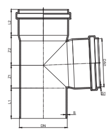
PLAIN BEND 45°

Size (in)	a	DN	S	Z1	Z2	L1	L2
4	45°	110	3.2	22.8	36.4	70	60
6	45°	160	4.0	33.1	50.2	95	80
8	45°	200	5.0	41.3	64.5	103.7	85

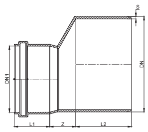
PLAIN BEND 87.5°

Size (in)	a	DN	S	Z1	Z2	L1	L2
4	87.5	110	3.2	52.7	66.3	70	60
6	87.5	160	4.0	76.5	93.7	95	80
8	87.5	200	5.0	95.8	118.8	104.8	85

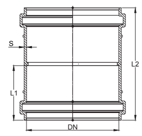
SINGLE TEE PLAIN

Size (in)	DN	DN1	S	Z1	Z2	L1	L2
4	110	111.0	3.2	52.7	69.3	70.0	60
6	160	161.5	4.1	76.6	98.7	95.0	80
8	200	201.8	5.0	96.7	125.2	103.7	85

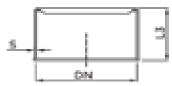
REDUCING COUPLER

Size (in)	DN	DN1	S	L1	L2	Z
6x4	160	110	4	60	94	43.9
8x6	200	161.5	5	70.9	100	43.1

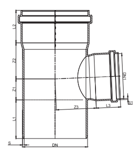
COUPLER

Size (in)	DN	S	L1	L2
4	110	2.8	65.5	135
6	160	3.6	90.0	185
8	200	4.5	95.7	197

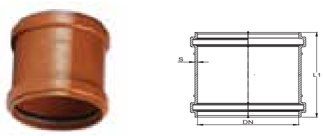
DUMMY PLUG

Size (in)	DN	S	L2
4	110	30	695
6	160	30	885

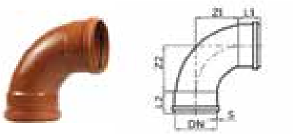
REDUCING SINGLE TEE

Size (in)	DN	DN1	S	S1	Z1	Z2	Z3	L1	L2	L3
6x4	160	111	4	2.8	51.6	88.7	98.2	95	8	54



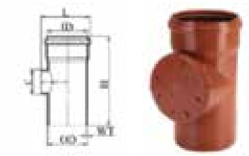
REPAIR COUPLER

Size (in)	DN	S	L1
4	110	2.8	135
6	160	3.6	185
8	200	4.5	197



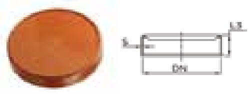
LONG SWEEP BEND

Size (in)	a	DN	S	Z1	Z2	L1	L2
4	45	110	4.1	76.7	66.7	64	54
6	45	160	4.1	76.7	93.7	87	70.9
6	45	200	5.0	95.7	118.9	110	85.0



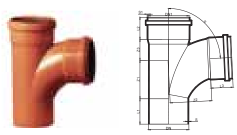
CLEANING PIPE

Size (in)	DN	ID	OD	WT	C	H	L
4	110	111.5	110	3.3	90.4	261.5	140.3
6	160	161.5	160	4.1	89.2	362.5	183.5
6	200	201.8	200	5.0	161.6	412.7	301.3



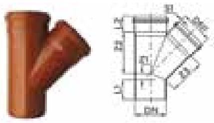
END CAP

Size (in)	DN	S	L2
8	200	3.1	38.7
10	250	3.6	40.5
12	315	4.1	52.5



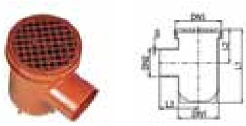
LONG SWEPT PLAIN

Size (in)	a	DN	DN1	S	S1	Z1	Z2	Z3	L1	L2	L3
4	87.5º	110	111	3.2	2.8	104.2	111.5	57	68.6	60	60
6	87.5º	160	161	4.0	3.6	119.3	159.8	81	92.3	80	80



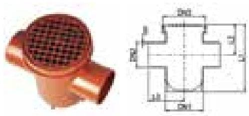
SINGLE WYE 45 PLAIN

Size (in)	DN	DN1	S	S1	Z1	Z2	Z3	L1	L2	L3
4	110	111.5	3.3	3.0	22.8	140	140	64	54	54
6	160	161.5	4.1	3.7	33.2	202.6	202.6	87	70.9	70.9
8	200	201.8	5.0	4.5	42	254.8	254.8	110	85	85



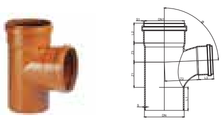
BOTTLE GULLY TRAP WITH JALI - SINGLE SIDE

Size (in)	a	DN	S	Z1	Z2	L1	L2
160 x 110	161	109.5	182	3.25	290	132	152.5



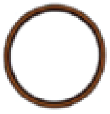
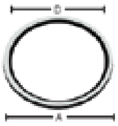
BOTTLE GULLY TRAP WITH JALI - DOUBLE SIDE

Size (in)	a	DN	S	Z1	Z2	L1	L2
160 x 110	161	109.5	182	3.25	290	132	152.5



REDUCING SWEPT TEE

Size (in)	a	DN	DN1	S	S1	Z1	Z1	Z3	L1	L2	L3
6x4	87.5	160	111	4	3.6	101.5	135.8	77	92.6	80	60

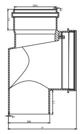



RUBBER WASHER

Size (in)	Size (mm)	ID	OD	WT
4	110	111.3	121.5	8.1
6	160	161.5	174.14	10.7
8	200	193.3	215.5	11.9
10	250	239.6	272.8	19.0
12	315	303	341.0	20.20

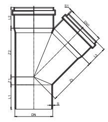
RUBBER LUBRICANTS

Size (in)	L1
100	Can
200	Can
500	Can

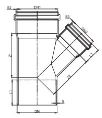
BACKFLOW PREVENTER

Size (in)	Size (mm)	Socket ID	Groove ID(A)	WT	H	L
4	110	111.5	121.3	3.2	170	280
6	160	161.5	175	4.0	257	396

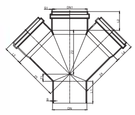
SINGLE WYE

Size (in)	DN	DN1	S	S	Z1	Z2	Z3	L1	L2	L3
4	110	111	3.2	2.8	22.7	140.2	140.2	70	60	60
6	160	161	4	3.6	33.1	202.6	202.6	95	80	80

REDUCING WYE

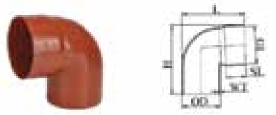
Size (in)	DN	DN1	S	S	Z1	Z2	Z3	L1	L2	L3
6x4	160	161.6	111	2.8	22.7	140.2	140.2	70	60	60

DOUBLE WYE

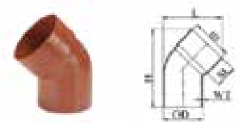
Size (in)	DN	DN1	S	S	Z1	Z2	Z3	L1	L2	L3
4	110	111	3.2	2.8	22.7	140.2	140.2	70	60	60
6	160	161	4	3.6	V33.1	202.6	202.6	95	80	80

SOLFIT UNDERGROUND FITTINGS



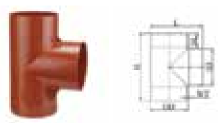
PLAIN BEND

Size (in)	Size (mm)	ID	OD	SL	WT	H	L
4	110	111.3	110.4	48	3.2	178	170
6	160	161.5	160.5	58	4.0	248.2	230



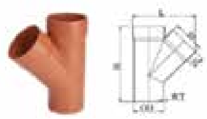
BEND 45º

Size (in)	Size (mm)	ID	OD	SL	WT	H	L
4	110	111.3	110.4	48	3.2	186	156
6	160	161.5	160.5	58	4.0	253	199



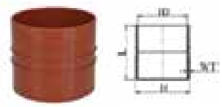
EQUAL TEE

Size (in)	Size (mm)	ID	OD	SL	WT	H	L
8	200	200.6	209.8	106	4.6	32	32



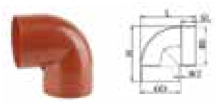
SINGLE Y

Size (in)	Size (mm)	ID	OD	SL	WT	H	L
4	110	111.3	110.4	48	3.2	276	235
6	160	161.5	160.5	58	4.0	380	330



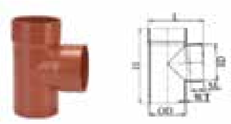
COUPLER

Size (in)	Size (mm)	ID	WT	H	L
4	110	111.3	3.2	120	102
6	160	161.5	4	170	125
8	200	200.6	4.6	210	219
10	250	250.8	5.7	262	271
12	315	316	7.2	331	338



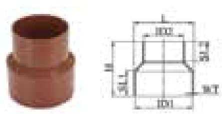
EQUAL ELBOW

Size (in)	Size (mm)	ID	OD	SL	WT	H	L
8	200	200.6	209.8	106	4.6	32	32



SINGLE TEE

Size (in)	Size (mm)	ID	OD	SL	WT	H	L
6	160	161.5	160.5	58	4.0	319	250

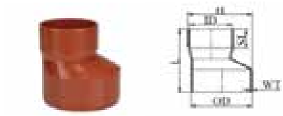


REDUCER COUPLER

Size (in)	Size (mm)	ID 1	ID 2	SL1	SL2	WT	H	L
6 x 4	160 x110	200.6	160.5	106	58	4.6	222	210
10 x 8	250 x 200	250.8	160.5	131	106	5.7	272	265
12 x 10	315 x 250	316.0	160.5	163.5	131	7.2	314	316

REDUCING TEE

Size (in)	Size (mm)	ID 1	ID 2	OD	SL 1	SL 2	WT	H	L
6 x 4	160 x 110	161.5	111.3	160.3	58	48	4	319	227



REDUCER OFFSET

Size (in)	Size (mm)	ID	OD	SL1	SL2	WT	H	L
6 x 4	160 x110	111.3	160.5	48	58	4	118.23	155.5

SOLVENT ADHESIVE

Weight in liters	Container Type
100	Can
250	Can
500	Can
1000	Can

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